Implementation of Dielectric ELC in ESPResSo

Bachelor's Thesis

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Simulation Technology B.Sc.

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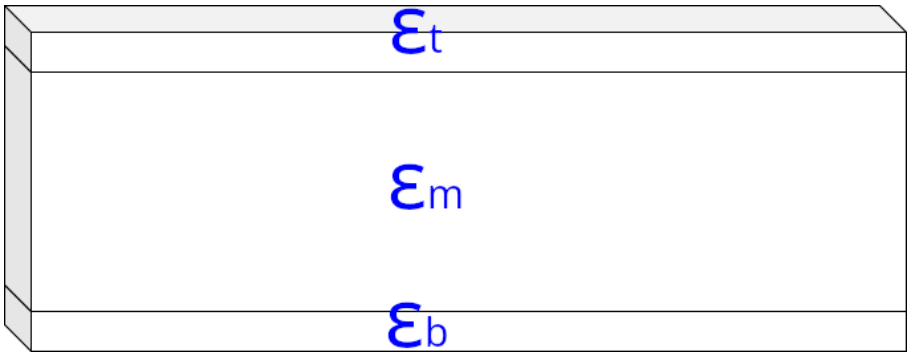
Motivation

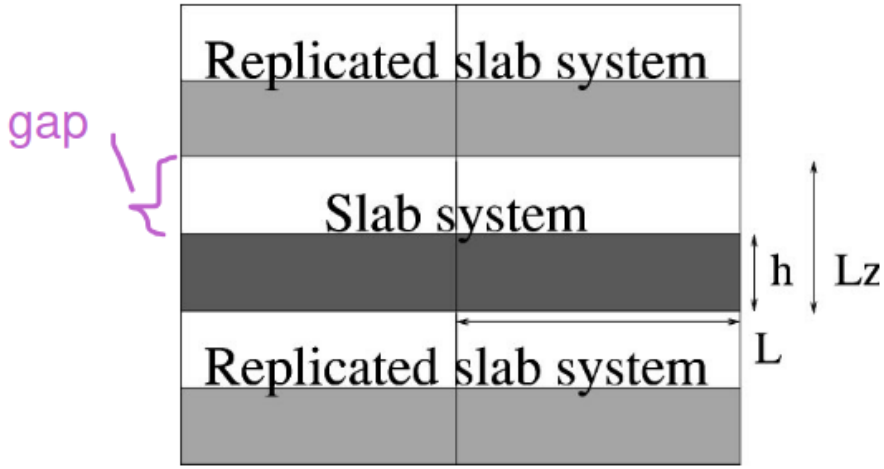
There already is an ELC implementation that can handle dielectric interfaces in ESPResSo, but:

- It has errors that are poorly understood.
- The code is unmaintainable and very difficult to read.

Goal: Replace the old ELC implementation with a clean, fixed, modern version.

Introduction





Developing in Python allows for faster iteration:

- Easier to develop, test, and prototype.
- Simplified visualization and validation of results.

Regular ELC: Development Strategy

Iterative/test-driven development by breaking the problem down:

- **Workflow per system:**
  - Compute reference solutions (analytical or brute-force).
  - Adjust ELC prototype to match reference solutions.
- **Systems analyzed:**
  - Large boxes (negligible PBC influence with $L_x, L_y >> 1$).
  - Smaller, neutral boxes (no non-neutral correction term yet).
  - Full parameter sweeps (Non-neutral, Madelung, random, and extreme parameters).

$$E_{2D+h} = E_{3D} + E_{\text{corr}} + E_{\text{non-neutral}} + E_{\text{dipole}}{}^{1}$$

[1] Tyagi, S., Arnold, A., & Holm, C. (2008). J. Chem. Phys. 129, 204102.

Regular ELC: Results

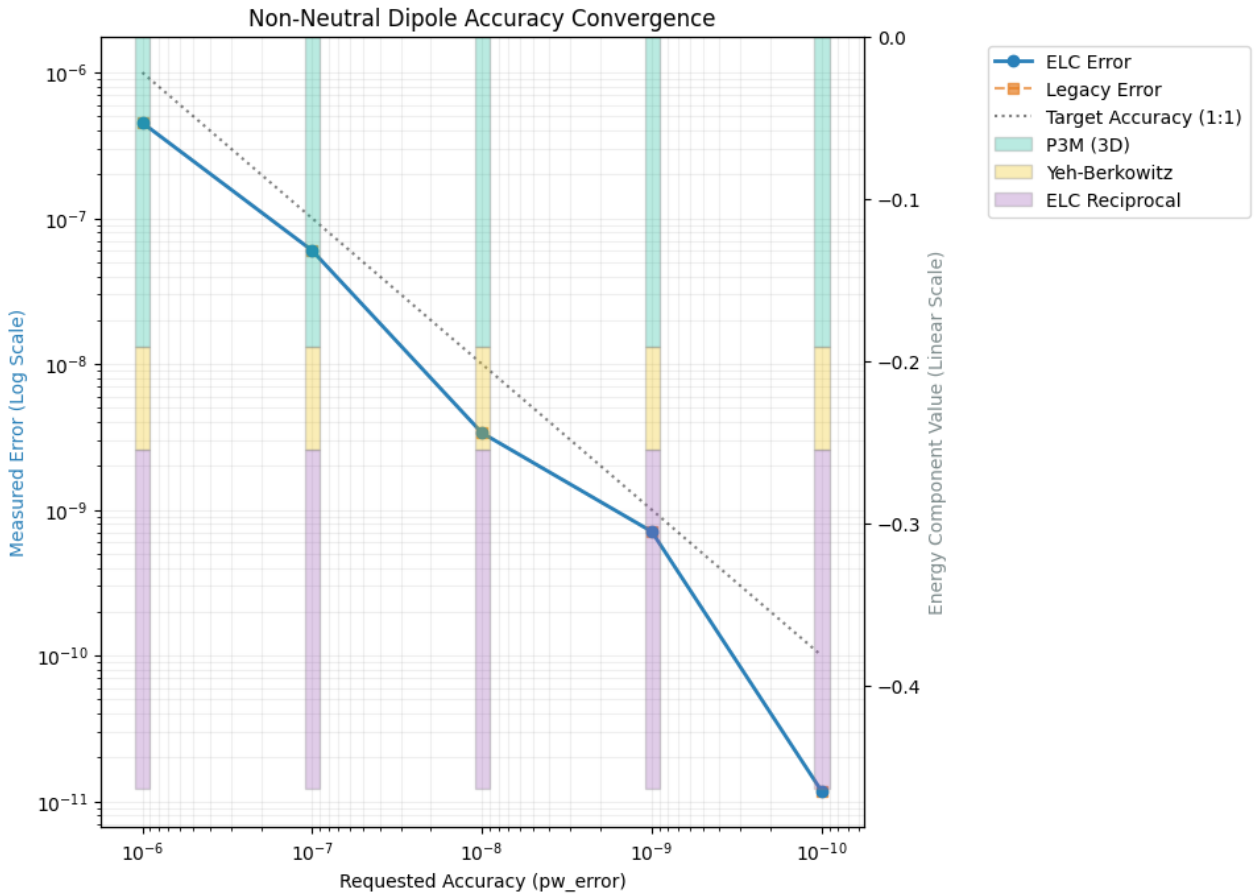
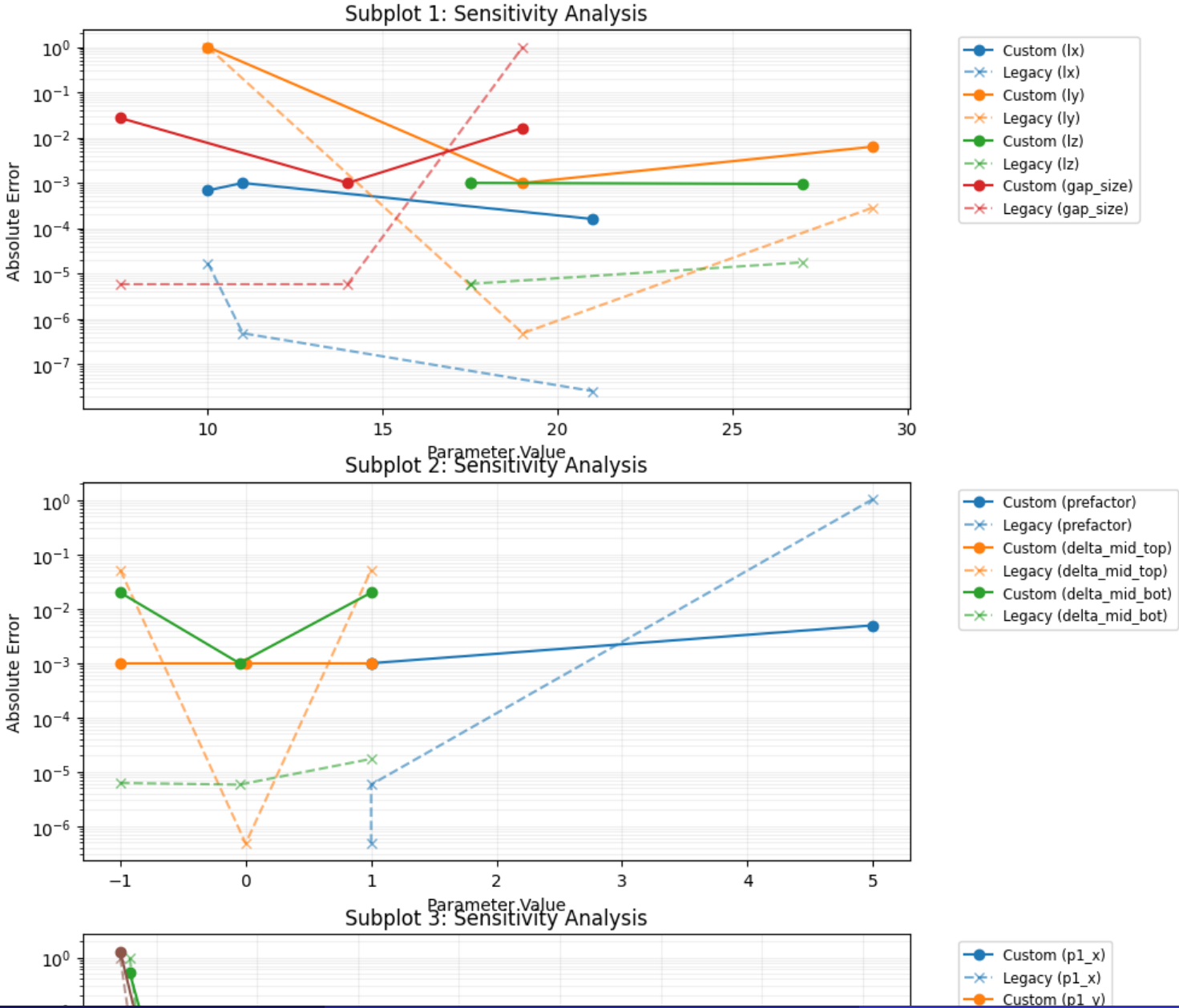


Figure: Validation of Regular ELC

Dielectric ELC: Single Plate (Current Status)

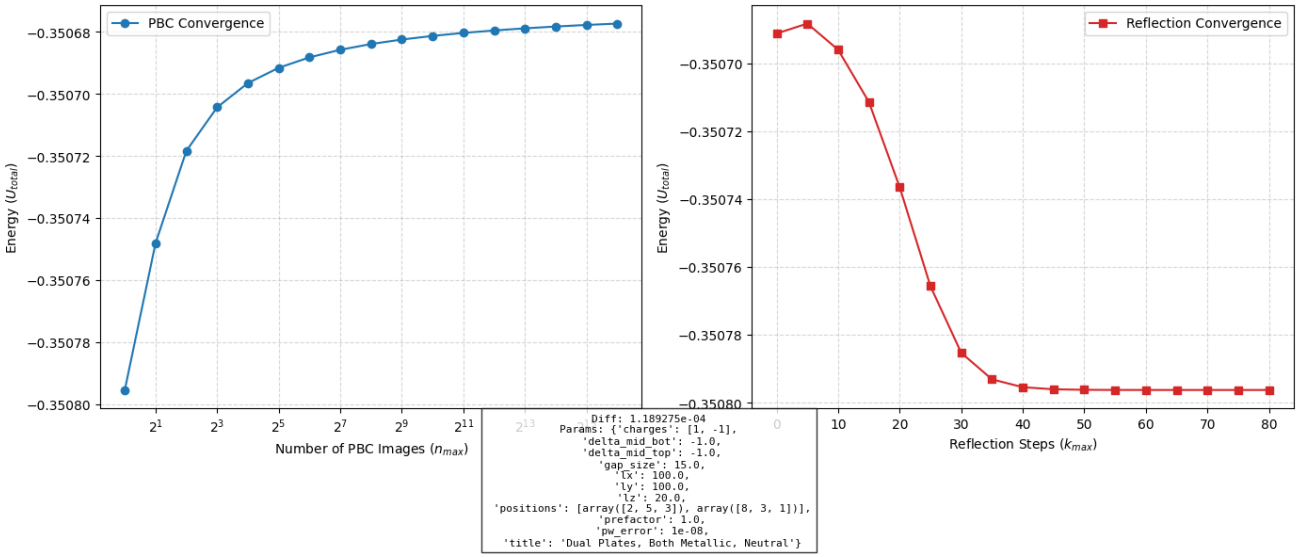
- Reference used: Ewald sum.
- Works for most parameters, though some outliers remain.

Dielectric ELC: Neighbourhood Scan



Dielectric ELC: Dual Plates

- Reference: Brute Force summation failed to converge.



Implementing in ESPResSo

- Upcoming: C++ implementation.
- Integration with **Kokkos** support for hardware acceleration.